

Hotel Management System

Python program to solve

the given question

Create class for hotel data.

class Hotel :

sortParam='name'

def __init__(self) -> None:

self.name=""

self.roomAvl=0

self.location=""

self.rating=int

self.pricePr=0

def __lt__(self,other):

getattr(self,Hotel.sortParam)<getattr(other,Hotel.sortParam)

Function to change sort parameter to

name

@classmethod

def sortByName(cls):

cls.sortParam='name'

Function to change sort parameter to

rating.

```
@classmethod
```

```
def sortByRate(cls):
```

```
    cls.sortParam='rating'
```

```
# Function to change sort parameter to
```

```
# room availability.
```

```
@classmethod
```

```
def sortByRoomAvailable(cls) :
```

```
    cls.sortParam='roomAvl'
```

```
def __repr__(self) -> str:
```

```
    return "PRHOTELS DATA:\nHotelName:{}\tRoom
```

```
Available:{}\tLocation:{}\tRating:{}\tPricePer
```

```
Room:{}".format(self.name,self.roomAvl,self.location,self.rating,self.pricePr)
```

```
# Create class for user data.
```

```
class User:
```

```
    def __init__(self) -> None:
```

```
        self.uname=""
```

```
        self.uld=0
```

```
        self.cost=0
```

```
    def __repr__(self) -> str:
```

```
        return "UserName:{}\tUserId:{}\tBooking
```

```
Cost:{}".format(self.uname,self.uld,self.cost)
```

Print hotels data.

```
def PrintHotelData(hotels):
```

```
    for h in hotels:
```

```
        print(h)
```

Sort Hotels data by name.

```
def SortHotelByName(hotels):
```

```
    print("SORT BY NAME:")
```

```
    Hotel.sortByName()
```

```
    hotels.sort()
```

```
    PrintHotelData(hotels)
```

```
    print()
```

Sort Hotels by rating

```
def SortHotelByRating(hotels):
```

```
    print("SORT BY A RATING:")
```

```
Hotel.sortByRate()
```

```
hotels.sort()
```

```
PrintHotelData(hotels)
```

```
print()
```

```
# Print Hotels for any city Location.
```

```
def PrintHotelBycity(s,hotels):
```

```
    print("HOTELS FOR {} LOCATION ARE:".format(s))
```

```
    hotelsByLoc=[h for h in hotels if h.location==s]
```

```
    PrintHotelData(hotelsByLoc)
```

```
    print()
```

```
# Sort hotels by room Available.
```

```
def SortByRoomAvailable(hotels):
```

```
    print("SORT BY ROOM AVAILABLE:")
```

```
    Hotel.sortByRoomAvailable()
```

```
    hotels.sort()
```

```
    PrintHotelData(hotels)
```

```
    print()
```

Print the user's data

```
def PrintUserData(userName, userId, bookingCost, hotels):
```

```
    users=[]
```

```
    # Access user data.
```

```
    for i in range(3) :
```

```
        u=User()
```

```
        u.uname = userName[i]
```

```
        u.uld = userId[i]
```

```
        u.cost = bookingCost[i]
```

```
        users.append(u)
```

```
    for i in range(len(users)) :
```

```
        print(users[i],"\tHotel name:",hotels[i].name)
```

Function to solve

Hotel Management problem

```
def HotelManagement(userName,
```

```
                    userId,
```

```
                    hotelName,
```

```
                    bookingCost,
```

```
                    rooms,
```

```
                    locations,
```

```
        ratings,  
        prices):
```

```
# Initialize arrays that stores
```

```
# hotel data and user data
```

```
hotels=[]
```

```
# Create Objects for
```

```
# hotel and user.
```

```
# Initialise the data
```

```
for i in range(3) :
```

```
    h=Hotel()
```

```
    h.name = hotelName[i]
```

```
    h.roomAvl = rooms[i]
```

```
    h.location = locations[i]
```

```
    h.rating = ratings[i]
```

```
    h.pricePr = prices[i]
```

```
    hotels.append(h)
```

```
print()
```

```
# Call the various operations
```

```
PrintHotelData(hotels)
```

```
SortHotelByName(hotels)
```

```
SortHotelByRating(hotels)
```

```
PrintHotelBycity("Bangalore",  
                hotels)  
SortByRoomAvailable(hotels)  
PrintUserData(userName,  
              userId,  
              bookingCost,  
              hotels)
```

Driver Code.

```
if __name__ == '__main__':
```

```
    # Initialize variables to stores
```

```
    # hotels data and user data.
```

```
    userName = ["U1", "U2", "U3"]
```

```
    userId = [2, 3, 4]
```

```
    hotelName = ["H1", "H2", "H3"]
```

```
    bookingCost = [1000, 1200, 1100]
```

```
    rooms = [4, 5, 6]
```

```
    locations = ["Bangalore",  
                "Bangalore",  
                "Mumbai"]
```

```
    ratings = [5, 5, 3]
```

```
    prices = [100, 200, 100]
```

Function to perform operations

HotelManagement(userName, userId,
hotelName, bookingCost,
rooms, locations,
ratings, prices)

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